

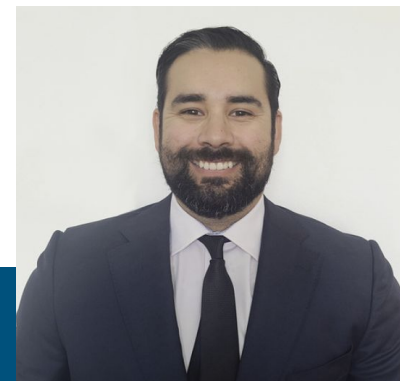
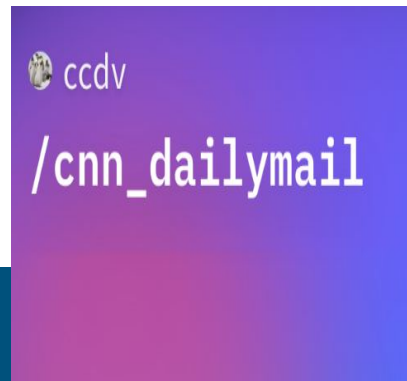
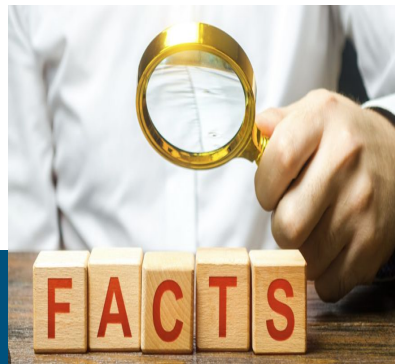
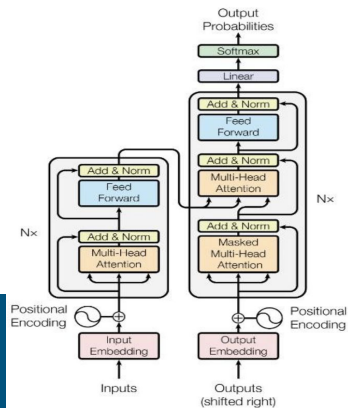
# When Metrics Say “Yes” But Humans Say “No”

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*A Case Study of Hallucination Reduction for Factual  
Summarization*

Author: Ryan Powers

# Project Characters



BART(Generator)

Encoder-decoder,  
140M params

FactCC(Verifier)

Binary classifier  
(Consistent/Inconsistent)

CNN/Daily Mail  
(Dataset)

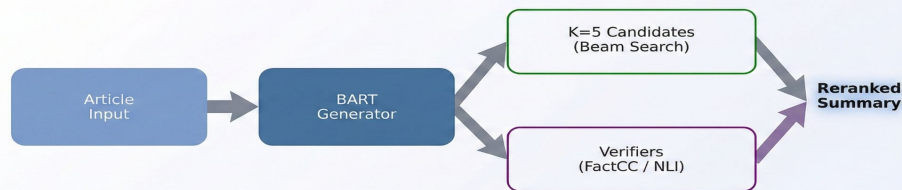
287K train / 13K val /  
11K test

Single Human  
Annotator

Blind comparison,  
stratified sample

# Task/Problem/Goal

Project Architecture: Verifier-Reranking Pipeline



*Can I just download a standard fact-checker and use it to fix my model's errors?*

## Task:

*Test whether better automatic metrics  $\Rightarrow$  better human-judged summaries.*

## Problem:

Summaries are fluent but sometimes factually wrong (hallucinations)

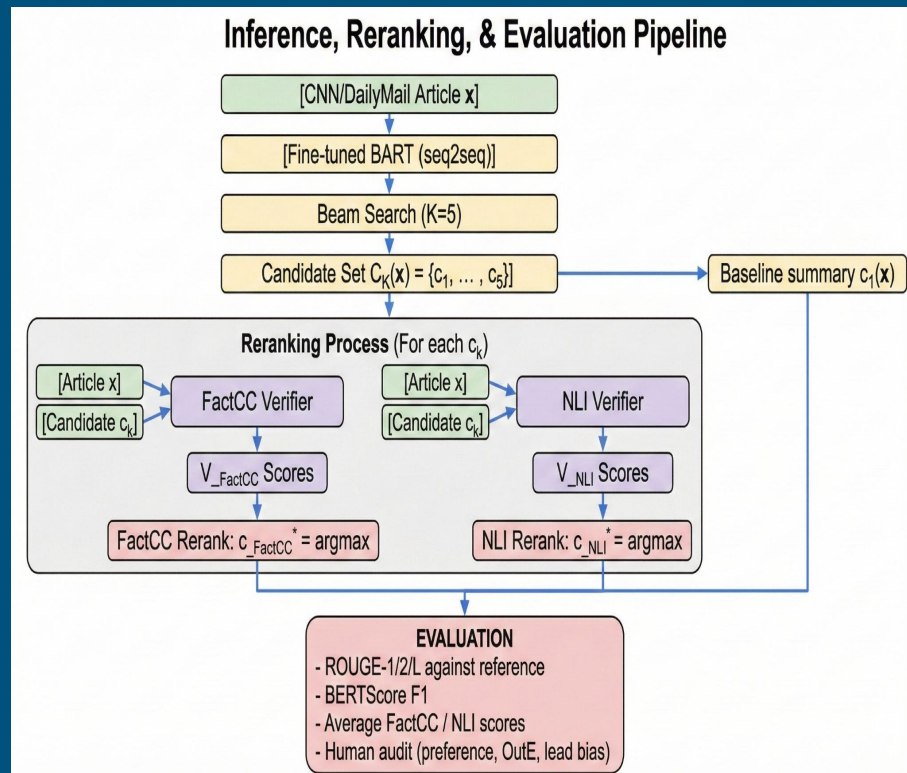
## Goal:

Improve factual consistency **without** retraining a bigger model, use reranking instead.

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# Architecture Diagram (Generate → Verify)

- Generate **K=5** candidates with BART (beam search)
- Score each candidate with:
  - **FactCC**: probability the summary is consistent with the article.
  - **RoBERTa-MNLI**: entailment score (for ablation).
- Pick the candidate with highest FactCC score.



# BART + FactCC / Experimental Setup

## Models

- **Generator:** facebook/bart-base (fine-tuned on 20k CNN/DM, 3 epochs, lr=5e-5) -
  - a. **Verifier A:** FactCC (factual consistency)
  - b. **Verifier B:** RoBERTa-MNLI (ablation)

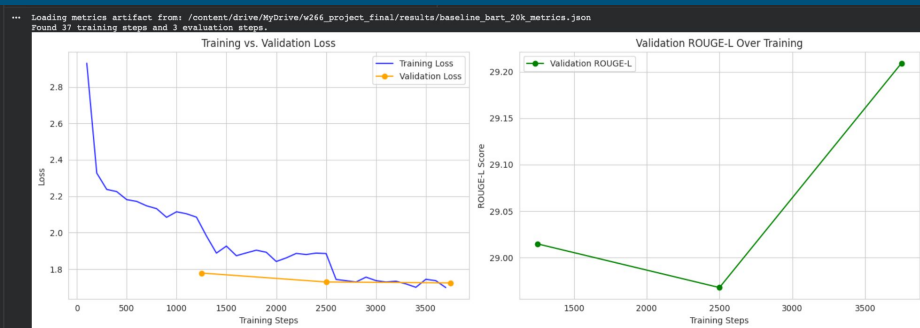
## Data

- **Train:** 20k subset (seed=42) - **Val:** 2k | **Test:** 11,490

## Decoding

- **Beam search, K=5, max length 140**

**Success Criteria** ✓  $\Delta\text{FactCC} \geq +2.0$  points ✓  
**ROUGE-L drop**  $\leq 1.0$  point



| Epoch | Training Loss | Validation Loss | Rouge1    | Rouge2    | RougeL    | RougeLsum |
|-------|---------------|-----------------|-----------|-----------|-----------|-----------|
| 1     | 2.084900      | 1.778065        | 42.660900 | 19.959000 | 29.014700 | 39.654000 |
| 2     | 1.885100      | 1.729170        | 42.443300 | 19.605700 | 28.967700 | 39.486900 |
| 3     | 1.698500      | 1.724090        | 42.618100 | 19.887200 | 29.209100 | 39.618100 |

## Tokenization Settings

| Parameter      | Value | Justification                                                    |
|----------------|-------|------------------------------------------------------------------|
| max_source_len | 1024  | Covers 95%+ of articles without truncation (from EDA)            |
| max_target_len | 128   | >2x the average summary length (~56 tokens); prevents truncation |

# Automatic Results

- **ROUGE-L:**
  - **LEAD-3: 24.9**
  - **BART baseline: 28.09**
  - **FactCC rerank: 27.92**
  - **NLI rerank: 28.08**
- **FactCC (avg factuality score):**
  - **Baseline: 0.427**
  - **FactCC rerank: 0.661**  
**(+55%,  $p < 0.00001$ )**
- **Success criteria satisfied:**
  - **$\Delta\text{FactCC} \geq +2.0$  ✓**
  - **ROUGE-L drop  $\leq 1.0$**   
**(-0.17) ✓**

Table 1: Main results on CNN/DailyMail test set ( $N=11,490$ ). R-1/2/L = ROUGE  $F_1$  ( $\times 100$ ). BS = BERTScore  $F_1$ . FactCC = mean factuality score. Best factuality in **bold**.

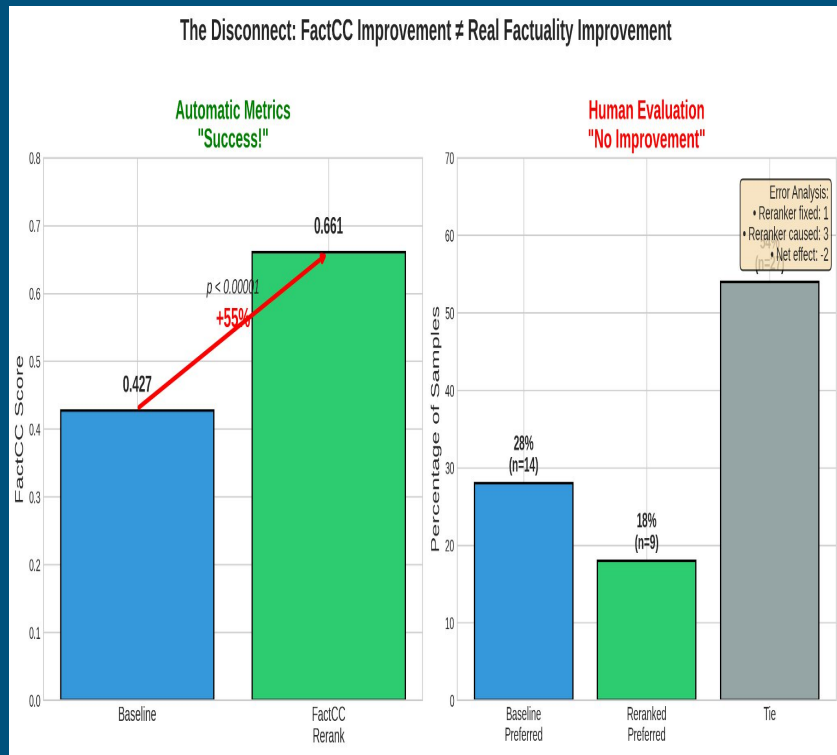
| System        | R-1   | R-2   | R-L   | BS    | FactCC       |
|---------------|-------|-------|-------|-------|--------------|
| LEAD-3        | 39.91 | 17.38 | 24.91 | —     | —            |
| BART Baseline | 41.12 | 18.56 | 28.09 | 0.882 | 0.427        |
| FactCC Rerank | 40.74 | 18.36 | 27.92 | 0.882 | <b>0.661</b> |
| NLI Rerank    | 40.78 | 18.48 | 28.08 | —     | —            |

$\Delta$  (FactCC Rerank vs Baseline): FactCC +0.234 (+55%,  $p < 0.0001$ ), R-L -0.17 (n.s.)

Table 2: Impact of candidate pool size ( $K$ ) on factuality. Marginal gain shows diminishing returns beyond  $K=3$ .

| $K$ | Avg FactCC | $\Delta$ from $K=1$ | Marginal Gain |
|-----|------------|---------------------|---------------|
| 1   | 0.427      | —                   | —             |
| 2   | 0.515      | +0.088              | +0.088        |
| 3   | 0.577      | +0.150              | +0.062        |
| 4   | 0.625      | +0.198              | +0.048        |
| 5   | 0.661      | +0.234              | +0.036        |

# Human Audit: 50 - Example Evaluation



- Stratified Sampling:
  - 30 where baseline vs rerank disagree
  - 20 where they agree
- Human (Single Annotator) preference:
  - Tie: 27 (54%)
  - Baseline preferred: 14 (28%)
  - Reranked preferred: 9 (18%)
- Net hallucination effect: reranker fixes 1 error, introduces 3 → **Net -2**

## Error Taxonomy (Human-Labeled, $N = 50$ )

| Error Type            | Definition                                                | Count (N=50)    |
|-----------------------|-----------------------------------------------------------|-----------------|
| <b>None</b>           | Summary contains no factual errors                        | <b>34 (68%)</b> |
| <b>Entity</b>         | Wrong name, number, place, or factual attribute           | 0 (0%)          |
| <b>Temporal</b>       | Incorrect time, date, or sequence of events               | 0 (0%)          |
| <b>Contradiction</b>  | Summary directly contradicts the article                  | 0 (0%)          |
| <b>Out-of-Context</b> | Fabricated or unsupported factual claim ("hallucination") | <b>16 (32%)</b> |



# When Reranking Helps vs Hurts

## The Success (Logic Repair)

- **Context:** Refugees formed a human chain to stop attackers.
- **Baseline (The Mistake):** "15 men arrested... But were *prevented* [from being arrested] because they huddled together..."
  - Status: **Logical Error** (Implies the chain stopped the police).
- **Reranked (The Fix):** "...claimed the men tried to throw other Christians off... but were *prevented* [from doing so] because they huddled together..."
  - Status: **Corrected** (The verifier caught the causal mismatch).

## The Failure (Hallucination)

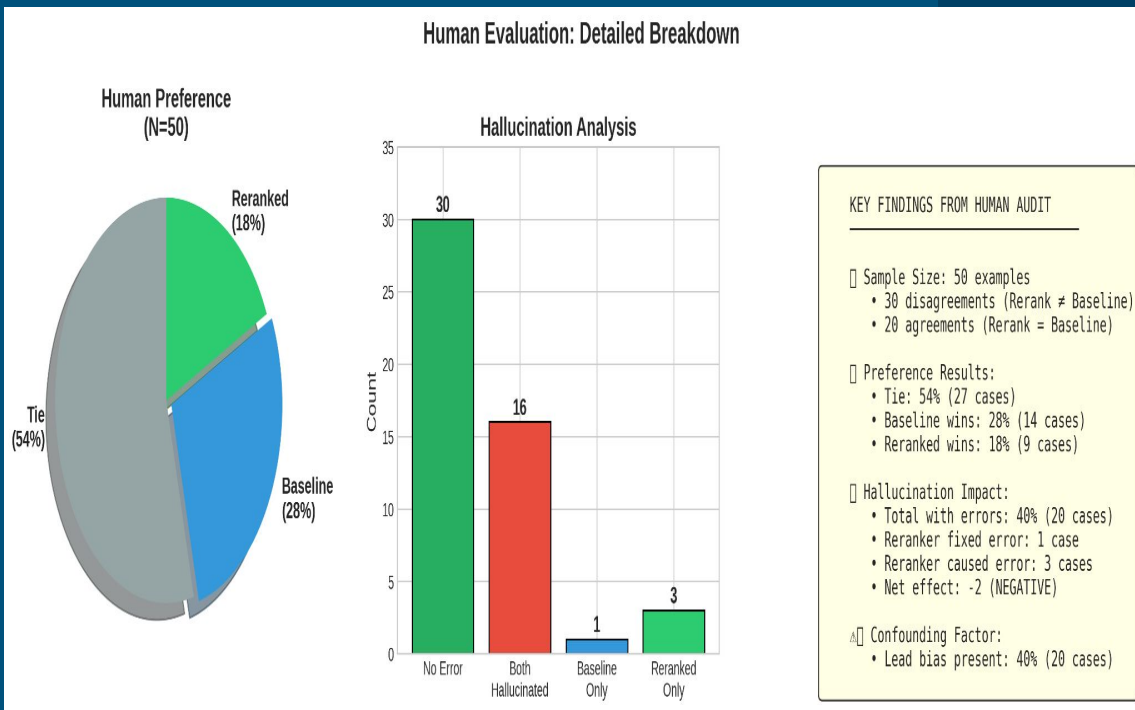
- **Context:** A father hit his toddler; store owner gave video to police.
- **Baseline (The Safe Bet):** "Justin Whittington... was taken into custody... relationship not yet revealed..."
  - Status: **Boring but Factual** (Stuck to the text).
- **Reranked (The Lie):** "...Police were notified... and asked the public for help identifying the man."
  - Status: **Hallucination** (The article never mentioned a public appeal).

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Verifier (FactCC) isn't actually checking facts it's just checking for patterns.

# Conclusion

*FactCC improved +55% ( $p < 0.00001$ ), but human audit showed 0% correction rate and net -2 hallucinations.*



**Low candidate diversity:** Beam search produces similar candidates (84% pairwise similarity)

**Verifier-human misalignment:** FactCC prefers extractive, lead-like summaries that humans find less informative

**Lead bias in CNN/DM:** LEAD-3 achieves 24.9 ROUGE-L only 3.2 points below BART so "safe" extractive choices score well

- **Why:** Goodhart's Law. Once I made FactCC the optimization target, the system learned to game FactCC rather than be more truthful.

# References & Github

- Lewis et al. 2020. “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation.” *ACL*.
- Hermann et al. 2015; Nallapati et al. 2016. CNN/DailyMail summarization dataset.
- Kryscinski et al. 2020. “Evaluating the Factual Consistency of Abstractive Text Summarization.” (FactCC).
- Ji et al. 2023. “Survey of Hallucination in Natural Language Generation.”
- Zha et al. 2023. “How Well Do Factuality Metrics Align with Human Judgments?” (for your QAGS/FactCC alignment discussion).

<https://github.com/rPowDS/MIDS266-NLP-final-project>